

概统第九次作业参考题解

2025.05.05

Q3. (1) 由于是简单随机抽样, 样本与总体分布一致: $E(X_i) = \mu$, $\text{Var}(X_i) = \sigma^2$. (2) 样本均值为 $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$, 由期望的线性性质得: $E(\bar{X}) = \mu$. 样本均值的方差为: $\text{Var}(\bar{X}) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i) + \frac{1}{n^2} \sum_{i \neq j} \text{Cov}(X_i, X_j)$. 对于无放回抽样: 方差 $\text{Var}(X_i) = \sigma^2$, 协方差 $\text{Cov}(X_i, X_j) = -\frac{\sigma^2}{N-1}$. 所以方差可以表示为: $\text{Var}(\bar{X}) = \frac{1}{n^2} \left(n\sigma^2 + n(n-1) \cdot \left(-\frac{\sigma^2}{N-1} \right) \right) = \frac{\sigma^2}{n} \left(\frac{N-n}{N-1} \right)$.

补充关于协方差 $\text{Cov}(X_i, X_j)$ 的推导:

$$\begin{aligned} \text{Cov}(X_i, X_j) &= E(X_i X_j) - E(X_i)E(X_j) \\ &= E(X_i X_j) - \mu^2 \\ &= \frac{1}{N(N-1)} \sum_{k=1}^N \sum_{l=1, l \neq k}^N x_k x_l - \mu^2 \\ &= \frac{1}{N(N-1)} \left(\left(\sum_{k=1}^N x_k \right)^2 - \sum_{k=1}^N x_k^2 \right) - \mu^2 \\ &= \frac{1}{N(N-1)} (N^2 \mu^2 - N \cdot (\mu^2 + \sigma^2)) - \mu^2 \\ &= -\frac{\sigma^2}{N-1}. \end{aligned}$$

Q4. (1) $\hat{p} = 1 - \frac{m_2}{a_1}$, $\hat{k} = \frac{a_1^2}{a_1 - m_2}$. (2) 样本量小时偏差较大; 无法保证 k 为整数; 无法保证 $k > \max_i X_i$; m_2 与 a_1 大小关系不确定.

Q5. 矩估计: $\hat{\theta} = \frac{2}{3}a_1$, MLE: $\hat{\theta} = \frac{1}{2} \max_i X_i$.

Q6. (1) 非负性显然, 积分验证正则性即可, 过程略. (2) $E(X) = a$. (利用正则性与奇函数性质); $\text{Var}(X) = 3\sigma^2$. (换元, 分部积分). 因此有 $\hat{a} = \bar{X}$, $\hat{\sigma}^2 = \frac{1}{3}m_2$. (3) $\ln L(a, \sigma) = -3n \ln(\sigma) + \sum_{i=1}^n 2 \ln(x_i - a) + \sum_{i=1}^n \left(-\frac{(x_i - a)^2}{2\sigma^2} \right) + \text{const.}$ 可利用 Newton-Rapson Method, Gradient Ascent 等求解.

Q7. 矩估计: $\hat{p} = \bar{X}$, MLE: $\hat{p} = \bar{X}$.

Q8. $\ln L(p_1, \dots, p_m) = \sum_{i=1}^m X_i \ln p_i$. 在 $\sum_{i=1}^m p_i = 1$ 的约束下极大化对数似然函数, 可采用 Lagrange Multiplier Method. MLE: $\hat{p}_i = \frac{X_i}{n}, \forall i \in$

$\{1, \dots, m\}$.

Q9. 矩估计: $\hat{\theta} = \frac{5}{6}$, MLE: $\hat{\theta} = \frac{5}{6}$.

Q10. 矩估计: $\hat{\theta} = \frac{\bar{X}}{1 - \bar{X}}$, MLE: $\hat{\theta} = \frac{-n}{\sum_{i=1}^n \ln X_i}$.